

From Collision Histories to Computable Closure: Proof-Guided Cumulants and Temporal Interaction Graphs

Working paper

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Abstract

Rigorous long-time kinetic limits propagate more than a smallness estimate for correlations: they retain structured, time-oriented information about the interaction histories that create those correlations. We formulate a computational program that converts this proof mechanism into measurable finite-scale closure diagnostics. The retained kinetic state is augmented by projected connected cumulants and compressed temporal interaction graphs, while the classical kinetic equation remains an explicit baseline. The principal test is whether this augmented state predicts one-layer closure defect and improves stable long-time rollout beyond models based only on the retained state or unordered cumulant magnitudes. This document is a paper scaffold; its claims remain hypotheses until the prescribed scaling, reversal, and ablation tests are completed.

1 Introduction

State the computational closure problem, the arrow-of-time obstruction, and the finite-scale research question.

2 Mathematical blueprint

Describe factorization, connected cumulants, time layering, partial expansion, collision-history molecules, canonical layered gardens, and hydrodynamic projection.

3 Computational closure state

For layer ℓ , use

$$S_\ell = (y_\ell, c_\ell, h_\ell, d_\ell, \sigma_\ell),$$

where y_ℓ is the retained state, c_ℓ contains projected cumulants, h_ℓ is a temporal history sketch, d_ℓ is closure defect, and σ_ℓ quantifies uncertainty.

4 Nonlinear waves to wave kinetics

Specify the cubic NLS ensemble, four-wave kinetic baseline, cumulant estimators, resonant hypergraph, and kinetic scaling sweep.

5 Arrow-of-time tests and ablations

Compare retained-state, cumulant-only, and oriented-history models on forward/reverse pairs and history-shuffled controls.

6 Hard-sphere transfer study

Specify event-driven dynamics, weak marginal/cumulant estimators, collision ancestry graphs, circuit rank, recollisions, and weak Boltzmann residuals.

7 Results

Report detectability, calibration, scaling, rollout stability, invariants, uncertainty, and cost. This section is intentionally empty before experiments.

8 Relation to the rigorous analysis

Separate exact theorem objects from finite-scale estimators and graph proxies. Explain why abstract cutting is used as a guide or certificate rather than as the physical time integrator.

9 Discussion

Discuss negative results, application transfer, and the next kinetic-to-fluid step.